

Tarun Mittal

Senior Software Engineer

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SUMMARY

- Senior Software Engineer with **5 years** of experience designing and owning cloud-native, high-throughput **distributed systems**.
- Expertise in **Python and Go** microservices, with applied experience in **AI-assisted** and **RAG** systems.
- Proven track record shipping production systems, leading backend migrations, and improving reliability, performance, and scalability.

SKILLS

- **Languages** – Python, Go (Golang), SQL, C, C++.
- **Backend & APIs** – FastAPI, Gin, Django, REST, gRPC, Pydantic, Celery, asyncio.
- **Generative AI** – LangGraph, LangChain, RAG, Milvus, pgvector, LLM Agents (OpenAI/Gemini/Claude), Tool Calling, MCP, Prompt Engineering, LangSmith, Offline Evaluation.
- **Data & Streaming** – Apache Kafka, Apache Flink, ClickHouse, BigQuery, ETL Pipelines.
- **Databases & Storage** – PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, DynamoDB, S3.
- **Cloud & DevOps** – GCP, AWS, Docker, Kubernetes, Bazel, CI/CD, Datadog, ELK, New Relic.
- **Architecture & Core Concepts** – Distributed Systems, Microservices, System Design (HLD/LLD), Data Structures & Algorithms, Caching, Reliability, Multithreading, Database Sharding, Concurrency, Idempotency.

PROFESSIONAL EXPERIENCE

Senior Software Engineer IMPACT ANALYTICS, Bangalore May 2026 – Present

AssortSmart — Senior Software Engineer (Platform & AI)

Agentic Flows & Orchestration

- Architected a **multi-pipeline AI merchandise decision engine (Keep/Drop, Missed Opportunities, Top Style)** evaluating **348k** article-seasons. Optimized inference at scale—processing **88k** items per pass—by blending deterministic KPI math with structured LLM invokes across **7 AI lenses**.
- Decoupled AI inference from a **2.11B-row** ClickHouse master by asynchronously serving context via JSON payloads rather than direct database querying. Engineered an eventually consistent write pattern via `ReplacingMergeTree` that guaranteed system availability by gracefully falling back to deterministic baselines during LLM timeouts.
- Engineered a **shared orchestration registry** to guarantee **88k-article** batch runs survive provider failures by implementing hard timeouts, circuit breakers, and durable checkpoints that bypass redundant deterministic scoring. Ensured full observability by stamping every output row with a frozen config hash and custom JSON telemetry tracking tokens, USD cost, step duration, and batch fallbacks.
- **Shipped “Ask Iris,”** a JWT-secured WebSocket copilot enabling planners to interactively query KPIs and inspect AI decisions on a multi-billion-row database. Prevented infinite LLM loops and multi-tenant data leaks by orchestrating a **LangGraph** supervisor with a 3-attempt evaluator loop, utilizing **LangSmith** for observability, and strictly freezing hierarchy access on the initial socket handshake.
- Built a **300-case** proxy evaluation harness and an **80%** accuracy CI gate to enforce strict deployment guardrails. Reduced live execution costs by **73%** at **100%** coverage by benchmarking models against a **74%** deterministic baseline, strategically freezing final decision weights when the LLM underperformed the baseline.

Core Infrastructure & Pipeline

- Architected a multi-tenant **Go/Gin** platform scaling to **10k peak RPS**, utilizing **Google Wire** for compile-time DI. Deployed as a self-protecting HTTP edge without a reverse proxy, guaranteeing high availability through native **h2c**, nested timeouts, and **Datadog** distributed tracing.
- Eliminated deployment bottlenecks by **architecting a dynamic KPI configurator** and custom formula parser. Engineered a real-time compilation layer that tokenizes operator-authored math into parameterized ClickHouse SQL fragments with native division-by-zero protection, **decoupling business logic from code releases**.
- Guaranteed strict **multi-tenant data isolation** and UAM-scoped hierarchy **access control** across **PostgreSQL** and **ClickHouse**. Secured system entry by implementing a Redis-fronted Firebase Admin to JWT/Google OIDC verification waterfall, constant-time API keys, and Postgres-backed role resolution.
- Reduced planner pivot latency by **15.5×** (**189s to 12s** on **250M-row** operations) by building a ClickHouse pre-aggregation layer for season and weekly rollups. Prevented **170GB** OOM crashes by slicing batch inserts into temporal chunks and enforcing strict distributed memory and disk-spill caps.
- Migrated ClickHouse rollups across Cloud clusters at **~344k rows/s** by writing a custom Go native-TLS data pump to bypass strict cross-cluster IP allowlist restrictions. Processed partitions up to **4.09B** rows using **500k-row** double-buffered batches, and eliminated partial-commit ghost rows using TSV ledgers and atomic partition rollbacks.
- **Tech** – Go, Gin, Python, FastAPI, LangGraph, ClickHouse, BigQuery, Redis, PostgreSQL, Datadog, LangSmith, GCP, Docker.

Financial Risk Management (FRM) Scoping Platform

- Owned the Financial Risk Management scoping backend, reducing quarterly reconciliation time by **70%** (from **14 days to 3 days**) against a **\$340M** materiality threshold. Engineered the Sheets-to-MySQL migration across **19M** raw GL rows, shipping **36 FastAPI** endpoints to automate custom financial math and a nested 4-level FSLI (L1–L4) hierarchy with dynamic quarter-annualization.
- Resolved ETL pipeline lock contention during financial close by designing an 8-table normalized MySQL schema on Uber’s SOADB featuring a polymorphic review-status model. Guaranteed real-time, multi-user collaboration integrity by implementing optimistic row-locking, atomic cross-table syncs, and deterministic SHA-256 natural-key hashing to eliminate data duplication across quarterly audit trails.
- Led **3 engineers** in a core SQLAlchemy 2.0 ORM migration that eliminated latent column-aliasing bugs and raised module statement coverage to **100%**. Orchestrated zero-downtime releases via stacked Bazel PRs, embedding reporting SQL allowlists and automated FSM gates with **50%** delta-variance checks to enforce SOX compliance.
- Tech** – Python, FastAPI, SQLAlchemy 2.0, Pydantic, MySQL, Bazel, Docker.

Menu Ingestion & Automation (Uber Eats)

- Accelerated restaurant onboarding from **24 hours to 2 hours** and saved **\$600K** annually by architecting an automated ingestion pipeline processing **30K+** menus monthly. Converted unstructured, multilingual PDFs into a strict catalog schema at **98%** field fidelity utilizing a **Gemini 2.5 Pro** and **LangChain RAG** pipeline over **Milvus** vector embeddings.
- Scaled menu extraction success to **95%+** by hardening a Selenium fleet with dynamic proxy pools and adaptive retries. Designed a **large-scale data ingestion and processing pipeline** using **Kafka** and **Apache Flink**, with stateful keyed deduplication and schema normalization to guarantee exactly-once catalog upserts.
- Tech** – Python, Selenium, Kafka, Flink, LangChain, Gemini, RAG, Milvus, GCP, Docker.

Document Compliance (Uber Mobility)

- Achieved **99.9%** regulatory compliance for Uber Mobility ANZ by automating driver and vehicle document verification workflows against local authority requirements. Eliminated **20 hours** of manual operational overhead per week by replacing manual review queues with a deterministic validation backend.

GST Compliance & E-Invoicing Platform

- Reduced p95 API latency by **75%** (from **1.2s to 300ms**) across **1,500+** enterprise clients by spearheading the migration of a legacy PHP monolith to a **FastAPI** microservices architecture. Mentored two engineers through the transition, optimizing backend design to drive overall system throughput from **700 to 4,000 requests per minute**.
- Architected a **high-concurrency** bulk e-invoicing pipeline processing **1M+** daily government submissions by orchestrating a **Kafka** event bus and a **PostgreSQL** database horizontally sharded by tax quarter. Enforced exactly-once, **idempotent** processing for **100K+** concurrent imports with **fault-tolerant state management** via dead-letter queues and bounded retries against volatile state IRP portals.
- Decreased incident triage time by **70%** and lowered database read load by **30%** by instrumenting **ELK** and **New Relic** distributed tracing alongside a Redis caching layer. Hardened the CI/CD release cycle by driving test coverage from **35% to 82%**, achieving a **98%** deployment success rate across the microservices ecosystem.
- Tech** – Python, FastAPI, Kafka, PostgreSQL, MongoDB, Redis, Elasticsearch, Celery, Docker, ELK, New Relic, AWS.

Courses Platform and Influencer Dashboard

- Stabilized the doubt-support platform to handle **10K+** daily queries and **10x** traffic spikes during live coding contests by migrating the backend to **Django**. Engineered high-throughput REST APIs for voting, pinning, and locking threads utilizing MySQL, MongoDB, Redis, and Elasticsearch, driving a **20%** lift in premium subscriptions.
- Accelerated platform course sales by **30%** by launching an influencer analytics and earnings dashboard to dynamically track transactions and affiliate coupon conversions. Increased engineering operational efficiency by **70%** by designing asynchronous scheduled workflows for automated video processing, user reminders, and data cleanup.
- Tech** – Python, Django, MySQL, MongoDB, Redis, Elasticsearch.

EDUCATION		
B.Tech (Information Technology)	Institute of Engineering & Technology, Lucknow	July 2017 – June 2021

ACHIEVEMENTS & CERTIFICATIONS

- Ranked **2260 / 37,000+** in the Google Code Jam Qualifier 2021 and finalist (top 3 nationally) at the **Smart India Hackathon 2020 (MHRD)**.
- Participated in the **Global AI Hackathon** at EPAM Systems.
- Earned **HackerRank Problem Solving** and **LangChain Academy** certifications ([HackerRank](#), [LangChain](#)).